ND-1021042

PROJECT APOLLO

LUNAR EXCURSION MODULE

PRIMARY GUIDANCE, NAVIGATION, AND CONTROL SYSTEM MANUAL

VOLUME II

AC ELECTRONICS
DIVISION OF GENERAL MOTORS
MILWAUKEE, WISCONSIN

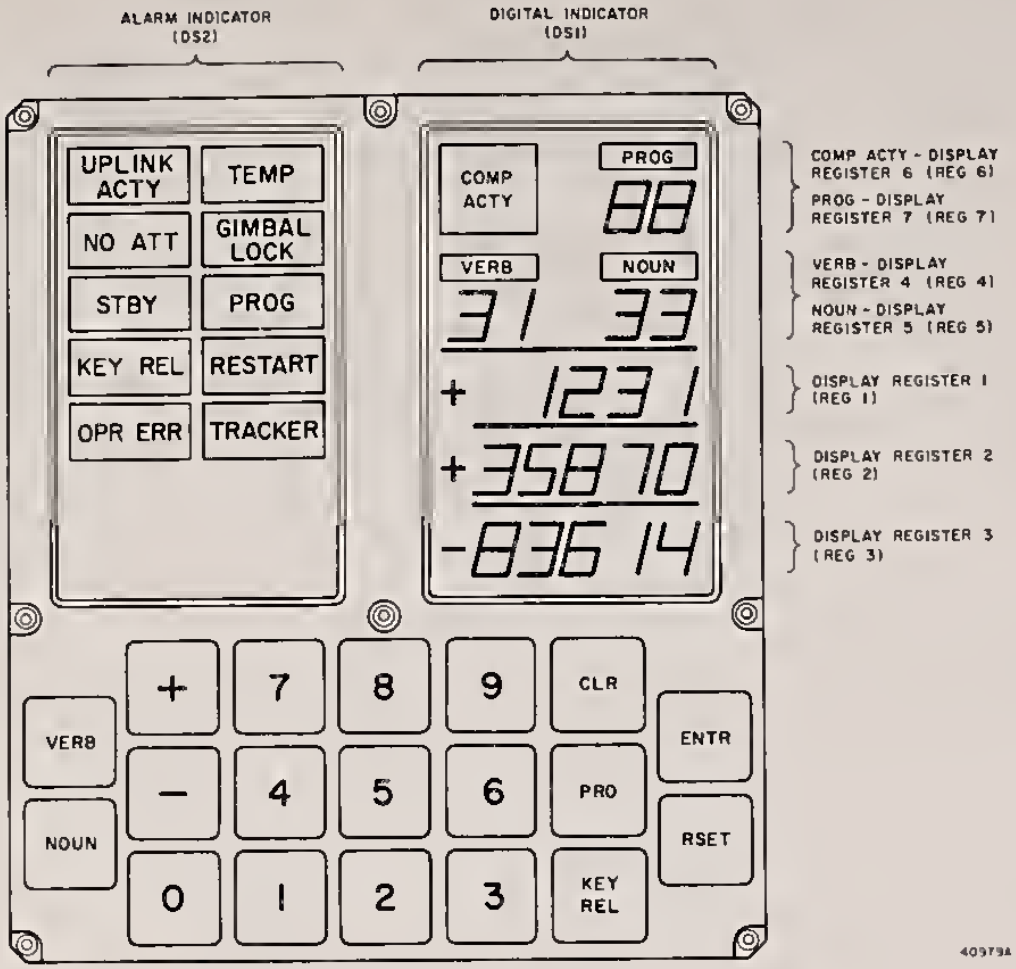


Figure 4-223. Keyboard and Display Front Panel

The key contacts (figure 4-224) are connected in series to ensure that only one keycode will be produced at one time. The binary keycode is produced by applying +28 volts through the key contacts to a diode network. The keycode initiates a program interruption (KEYRUPT) in the computer. When a key is released, signal KEYRST resets the input channel, thus clearing it for the acceptance of another keycode. A key must be released before another key is pressed to have information processed by the computer.

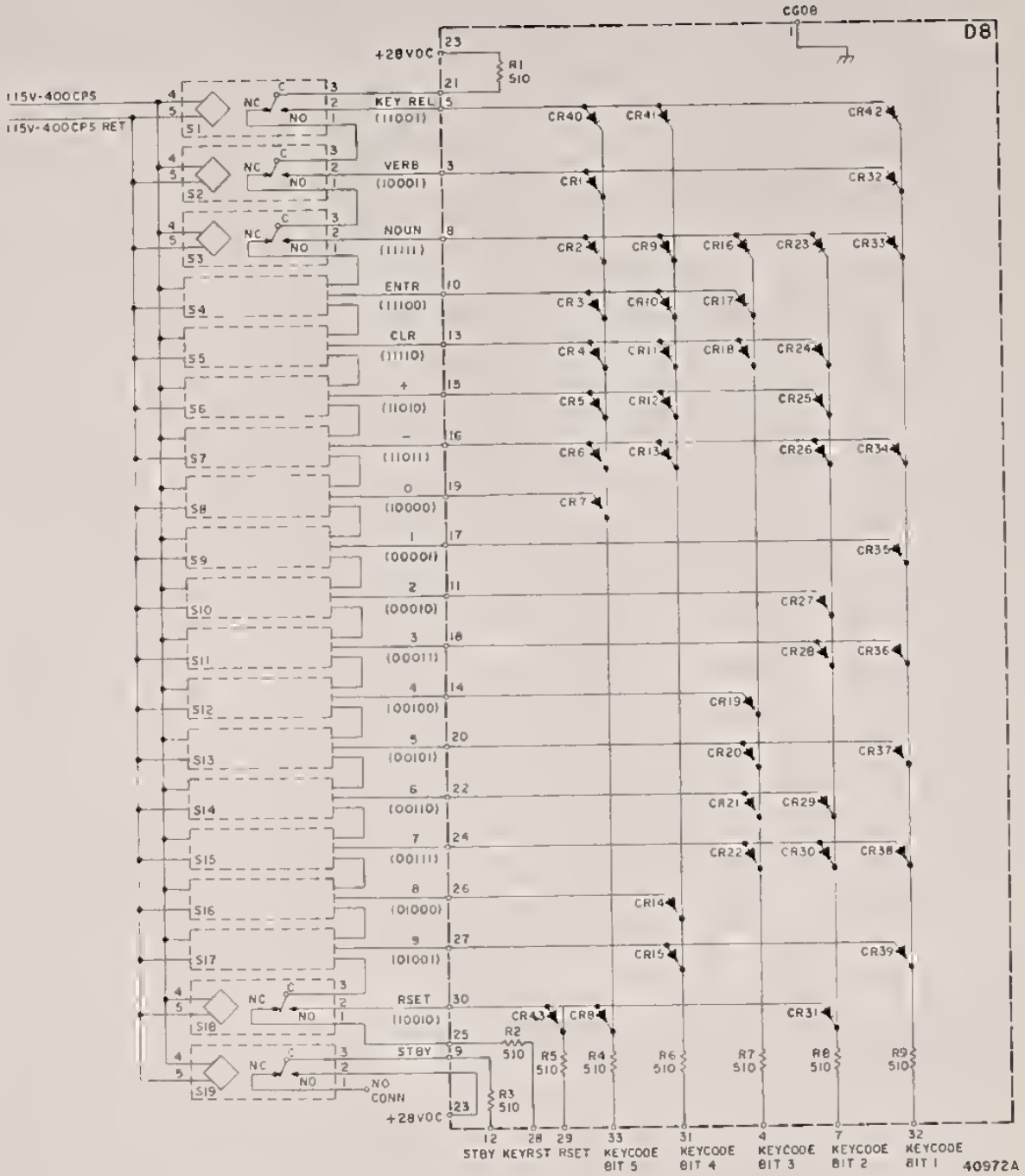


Figure 4-224. DSKY Keyboard Schematic Diagram

The display section (figure 4-223) contains 24 digit displays (21 for numerical and 3 for sign) and 15 indicators (spare included). The 24 digit displays (DS1 except COMP ACTY) and 15 indicators (DS2 plus COMP ACTY) are arranged as shown in figure 4-223. The displays and indicators are luminescent-coated-glass assemblies which glow when a voltage is applied to the coating. The displays (digit and sign) are segmented, and a display voltage of 250 vac from the DSKY power supply is applied to each segment through contacts in the relay matrix. The indicators are made in one piece. Except for COMP ACTY which receives 250 vac, all the indicators receive a display voltage of +5 volts from the spacecraft through relay contacts. The brightness of certain displays (digit, sign, and COMP ACTY) varies as a function of the voltage and frequency applied to the coating. The voltage can be varied using the brightness control on the astronauts control panel in the PGNCS. The operating controls and indicators, and their functions, are listed in table 3-VIII.

The standard procedure for communicating with the computer is to press seven keys in the following sequence: VERB-DIGIT-DIGIT, NOUN-DIGIT-DIGIT, and ENTR.

Pressing the VERB key on the keyboard clears the VERB displays on the display and indicators. The two digits punched in next are interpreted as a VERB code and displayed in the VERB section. This same operation occurs using the NOUN and two digits. The operation of the VERB-NOUN code is not initiated in the computer until key ENTR is pressed. If an error is noticed in either the VERB or NOUN before ENTR is pressed, it may be corrected by repunching either the VERB or NOUN key and the correct code.

If the VERB-NOUN combination punched in requires additional data to be furnished by the operator, the VERB and NOUN displays flash once every 1.5 seconds after the ENTR key has been pressed. The flashing indicates that the operator should punch-in the required data on the keyboard. After punching in the required data and the ENTR key, the flashing ceases.

Octal and decimal data words can be punched in. The computer assumes that an octal data word will be entered if a sign key (+ or -) is not pressed. If digit key 8 or 9 is pressed while loading an octal data word, indicator OPR ERR flashes once every 1.5 seconds. Whenever key (+) or key (-) is pressed, the corresponding signal is displayed and the computer assumes that a decimal word is to be entered. If an error is noticed while punching in either octal or decimal data, the CLR key can be pressed and the correct entry can be made if the ENTR key has not been pressed. All data words entered must be either octal or decimal; combinations of octal and decimal are not permitted.

To eliminate the flashing of indicator OPR ERR due to irregular keyboard entrys, key RSET must be pressed. In addition to the keycode, a hard wired signal (+28 vdc) is applied to the computer. Both the keycode and hard wire signal extinguish the status indicator OPR ERR as well as the five caution indicators: TEMP, GIMBAL LOCK, PROG, RESTART, and TRACKER. Thus, key RSET may also be used to test for the presence of a continuous caution rather than a transient caution condition.

4-5.10.2.2 Decoder. The decoder (figure 4-225) contains four relay word drivers (circuits 002 and 003 make up one driver), a diode matrix, and 12 row select drivers. The relay word drivers receive bits 15 through 12 of channel 10. Combinations of these

4 bits select 1 of 12 rows of relays in the relay matrix. The 12 code combinations from channel 10, are shown beside their particular row selection number on figure 4-225. For simplification only the selection of row 1 is discussed. The code for row 1 selection (0001) is inverted in the interface circuits (A25) and applied to DSKY connector J9 as signals CE228 through CE225. To identify row 1, signals CE228 through CE226 are logic ONE's and signal CE225 is a logic ZERO. A logic ONE shuts transistor Q1 off, which holds transistor Q2 off and allows transistor Q3 to conduct. Therefore, the X outputs of circuits controlled by signals CE228 through CE226 are logic ONE's, and the Y output of the circuit controlled by signal CE225 is a logic ONE.

The diode matrix receives the 8-bit output from the four relay word drivers. The matrix is wired so each 8-bit input produces a logic ONE on only one output line to the word drivers. For row 1 selection, diode CR53 of modules D2, D3, and D4, and diode CR63 of module D5 must be reverse biased. When these diodes are not conducting, row select driver circuit 004 on module D1 is activated. A current path is provided from +28 volts of the row selection driver voltage source (through transistor Q1, R8 of CKT 004, CR44, R9) to 0 vdc. Thus, parallel transistors Q4 and Q5 conduct and supply 0 vdc, representing row 1 selection, to the relay matrix.

The bottom row of diodes in the diode matrix (CR54 of modules D2 through D5) are used to detect the presence of logic ZERO's in bits 12 through 15 of channel 10. During normal operation at least one of the four diodes is forward biased and applies 0 vdc to the row selection driver voltage source. This 0 vdc is needed to supply a row selection signal (0 vdc) to the relay matrix as discussed previously. If the four most significant bits of channel 10 are ZERO's, all four diodes are reverse biased and +28 volts (+28 BCR) are applied to the input of the row selection driver voltage source. An input of +28 volts turns transistor Q2 on which in turn keeps transistor Q3 off. With no output from Q3, transistor Q1 is held off and the output of transistor Q1 is effectively an open circuit. Thus, there is no voltage source for circuit operation of any row selection driver circuit.

The indicator driver module circuits (figure 4-226) are used in making up the decoder, relay matrix, and status and caution circuits. They are introduced at this time to assist in better understanding the previously mentioned areas. All six indicator driver modules (D1 through D6) are identical and interchangeable.

4-5.10.2.3 Relay Matrix. The relay matrix (figure 4-227) consists of 11 relay bit drivers and 12 rows of latching relays.

Each relay bit driver accepts 1 of 11 bits (11 through 1) of channel 10. For simplification only bit 11 (circuit 006 on module D6) is discussed. When bit 11 of channel 10 is a ONE, it is inverted in the interface circuits (A25) and applied to DSKY connector J9 as signal CE224. A ZERO input to circuit 006 turns transistor Q8 on which switches transistor Q9 off. Thus, +28 volts is present on pin 10 (TURN-ON) of the column of relays dealing with the plus and minus sign display. With a row selection signal from the diode matrix (0 vdc) and a column select signal from a relay bit driver (+28 volts), a single relay within the relay matrix is controlled.

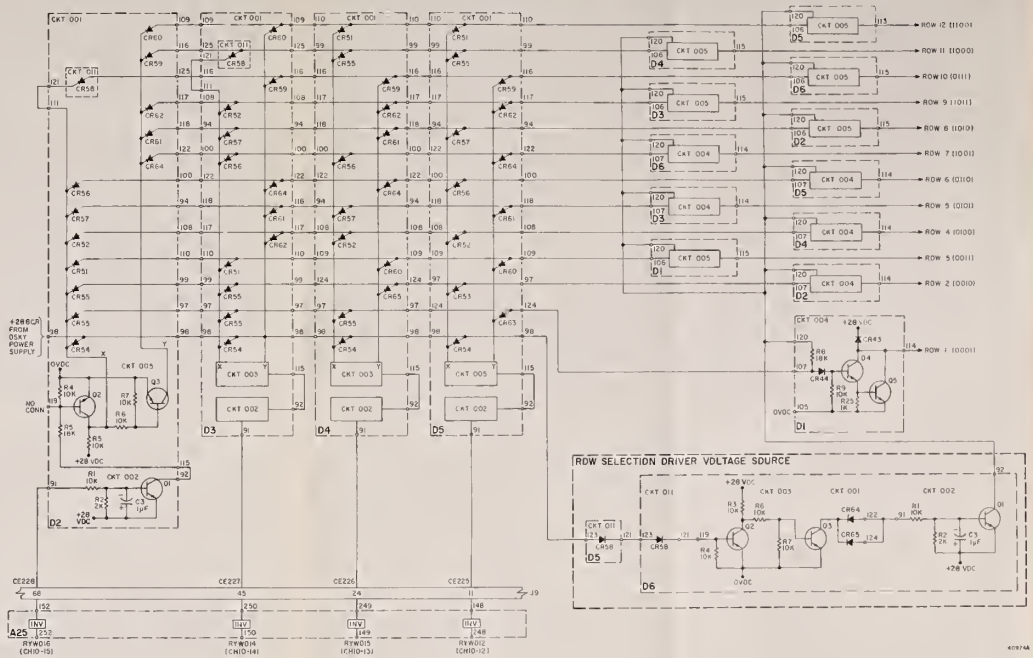


Figure 4-225. DSKY Decoder Schematic Diagram

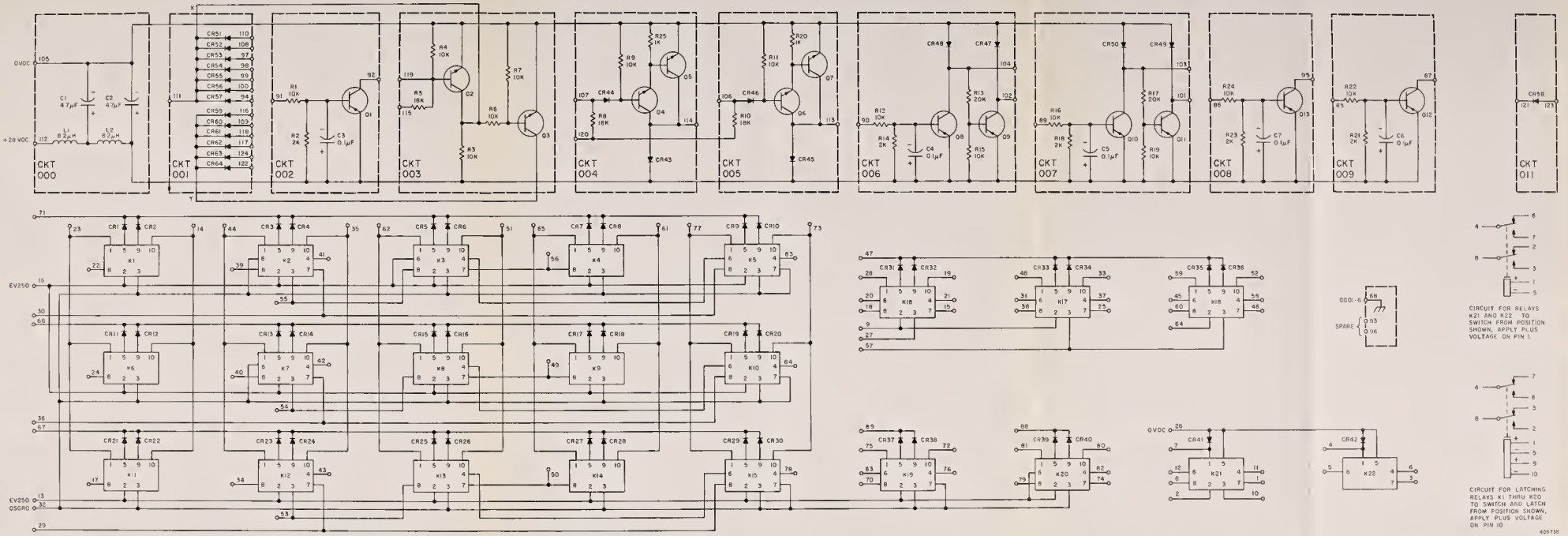


Figure 4-226. DSKY Indicator Driver Modules (D1-D6)

All of the relays in the relay matrix control the DSKY displays. Row 12 controls the indicators associated with the status and caution indicators (DS2) and, in addition, supplies a PGNS CAUTION signal to the display and control section of the PGNCS. Table 4-C relates the content of channel 10 to the row and column selected and the digit or indicator display controlled by the individual relay. Five relays are required to display one digit. Relay bit drivers 10 through 6 control the display of one digit and relay bit drivers 5 through 1 control the display of a second digit. Relay bit driver 11 causes the display of a plus or minus sign. The five-bit code necessary to display digits 0 through 9 in any display location is listed in table 4-CI. The relays representing REG3-POS1 of row 1 are used as an example. A logic ONE indicates that the relay is energized. For identification of display locations refer to figure 4-228.

Energizing the proper relays within the relay matrix (rows 1 through 11) allows approximately 250 vac from the DSKY power supply to be routed through the relay contacts to the various segments of the electroluminescent digit and sign indicators. Figure 4-229 illustrates the relays, their codes, and a display coding key.

Energizing a relay in row 12 allows +5 volts from the electrical power system in the spacecraft to be routed through the relay contacts to a status or caution indicator. Relays K16, K17, and K18 are spares. In addition, relays PROG, TRACKER, and GIMBAL LOCK receive the signal alarm common from the spacecraft and, when energized, supply signal PGNS CAUTION to the PGNCS.

Row Select	CHANNEL 10 BITS														
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
	RLY WD				RLY BITS										
1	0	0	0	1	R3 -	Reg 3		Pos 2			Reg 3		Pos 1		
2	0	0	1	0	R3 +	Reg 3		Pos 4			Reg 3		Pos 3		
3	0	0	1	1		Reg 3		Pos 1			Reg 3		Pos 5		
4	0	1	0	0	R2 -	Reg 2		Pos 3			Reg 2		Pos 2		
5	0	1	0	1	R2 +	Reg 2		Pos 5			Reg 2		Pos 4		
6	0	1	1	0	R1 -	Reg 1		Pos 2			Reg 1		Pos 1		
7	1	0	0	1		Noun		Pos 2			Noun		Pos 1		
8	1	0	1	0		Verb		Pos 2			Verb		Pos 1		
9	1	0	1	1		Program		Pos 1			Program		Pos 2		
10	0	1	1	1	R1 +	Reg 1		Pos 4			Reg 1		Pos 3		
11	1	0	0	0		Spares					Reg 1		Pos 5		
12	1	1	0	0			Program	Tracker	Spare	Gimbal Lock	Spare	No Att	Spare	Spare	Spare

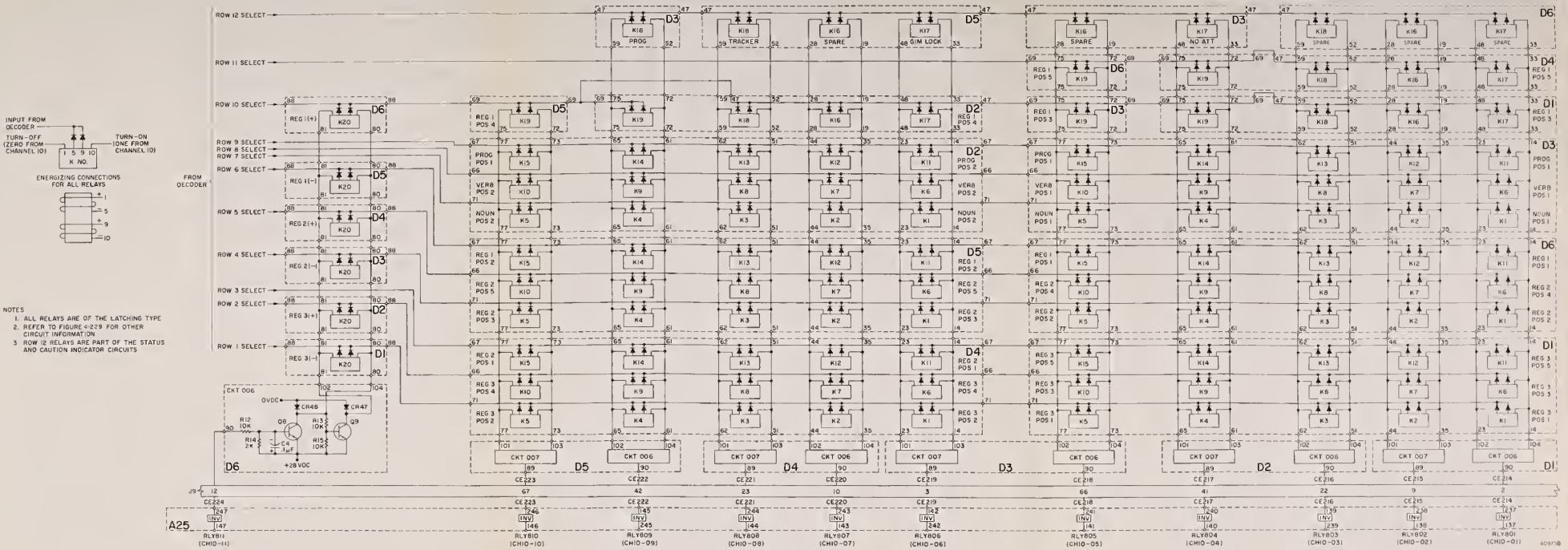
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Table 4-C. Relay Matrix Codes

Table 4-C1. Digit Code

Relays					Digit Displayed
K5	K4	K3	K2	K1	
0	0	0	0	0	Blank
1	0	1	0	1	0
0	0	0	1	1	1
1	1	0	0	1	2
1	1	0	1	1	3
0	1	1	1	1	4
1	1	1	1	0	5
1	1	1	0	0	6
1	0	0	1	1	7
1	1	1	0	1	8
1	1	1	1	1	9

4-5.10.2.4 Status and Caution Circuits. The status and caution circuits for the LGC (figure 4-230) consist of driver circuits and associated non-latching relays. For simplification only circuit 008 on module D4 is discussed. When signal ISS WARNING is a logic ZERO it will turn on transistor Q13 and supply +28 volts to associated relay K21. Relay K21 energizes and routes input signal ALARM COMMON through its contacts to the display and control section of the PGNCS as signal ISS WARNING. The driver circuits and relays associated with signals LGC WARNING, TEMP CAUTION, and RESTART also receive signal ALARM COMMON. Signal TEMP CAUTION or RESTART causes the generation of signal PGNS CAUTION which is applied to the PGNCS and also causes +5 volt caution power to be applied to the respective indicators on the DSKY front panel.



NOTES

1. ALL RELAYS ARE OF THE LATCHING TYPE
2. REFER TO FIGURE 4-229 FOR OTHER CIRCUIT INFORMATION
3. ROW 12 RELAYS ARE PART OF THE STATUS AND CAUTION INDICATOR CIRCUITS

Figure 4-227. Relay Matrix Schematic Diagram

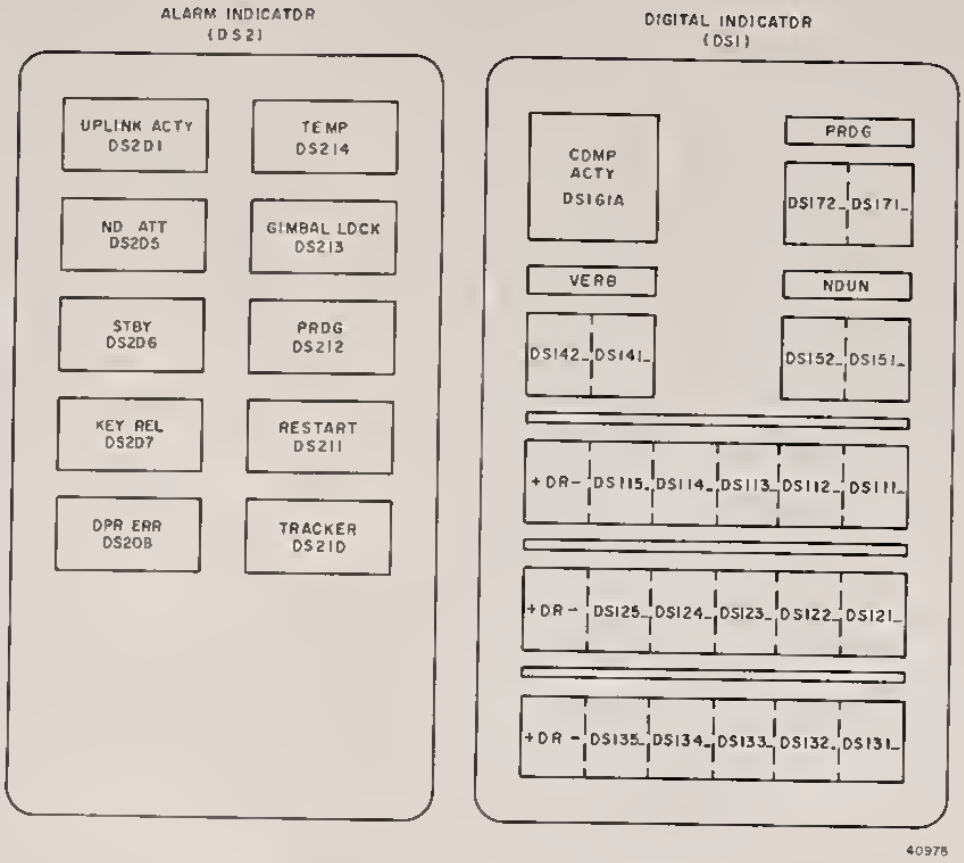


Figure 4-228. DSKY Display Locations

The driver circuits and relays associated with signals UPLINK ACTY, OPR ERROR, KEY REL, and STBY will apply, when activated, +5 volt status power to their respective indicators on the DSKY front panel. Indicators OPR ERR and KEY REL flash at a 1.5 cps rate. Indicator COMP ACTY is physically a part of digital indicator DSI. However, electrically it is part of the status circuits. When signal COMP ACTY is present, 250 vac is routed through the relay contacts to its indicator. Circuits dealing with signals RR ENABLE LOC ON, LGC WARNING, LRDR POS CMD, and ISS WARNING do not have visual indications on the front panel of the DSKY.

The verb-noun flash causes the verb-noun indicators to flash by interrupting, at a 1.5 cps rate, what normally is 250 vac being applied to the verb-noun relays in the relay matrix.

The status and caution circuits for the CMC are illustrated on figure 4-231. These circuits are identical in operation to those in the LGC. Several interface differences exist, however. These can be determined by examining the inputs and outputs to the status and caution circuits for the CMC on figure 4-231.

4-5.10.2.5 Power Supply. The DSKY power supply (figure 4-232) utilizes +28 vdc and +14VSW from the computer power supply, and 800 cps from the timer to generate a display voltage of approximately 250 vac, 800 cps. The power supply contains three transformer-coupled, push-pull amplifiers. The input to the first stage is an 800-cps square wave varying about a +14 vdc level. The dc level is controlled by the brightness control on the astronauts' control panel. Transformers T1 and T2 step up the voltage applied to their primary windings. The output from the third push-pull stage is applied to saturable reactor L2.

Reactor L2 and its associated circuit regulate the voltage applied to the displays. The displays act as a variable capacitive load that varies as a function of the number of indicators that are on. Changes in the load are reflected back to the control winding of L2 through the full-wave bridge rectifier, CR1 through CR4. As the number of indicators which are on increases, the voltage applied to the control winding is increased. An increase in voltage through the control winding drives reactor L2 further into saturation and keeps the output relatively constant. If the load decreases, the voltage through the control winding decreases, and L2 is less-saturated.

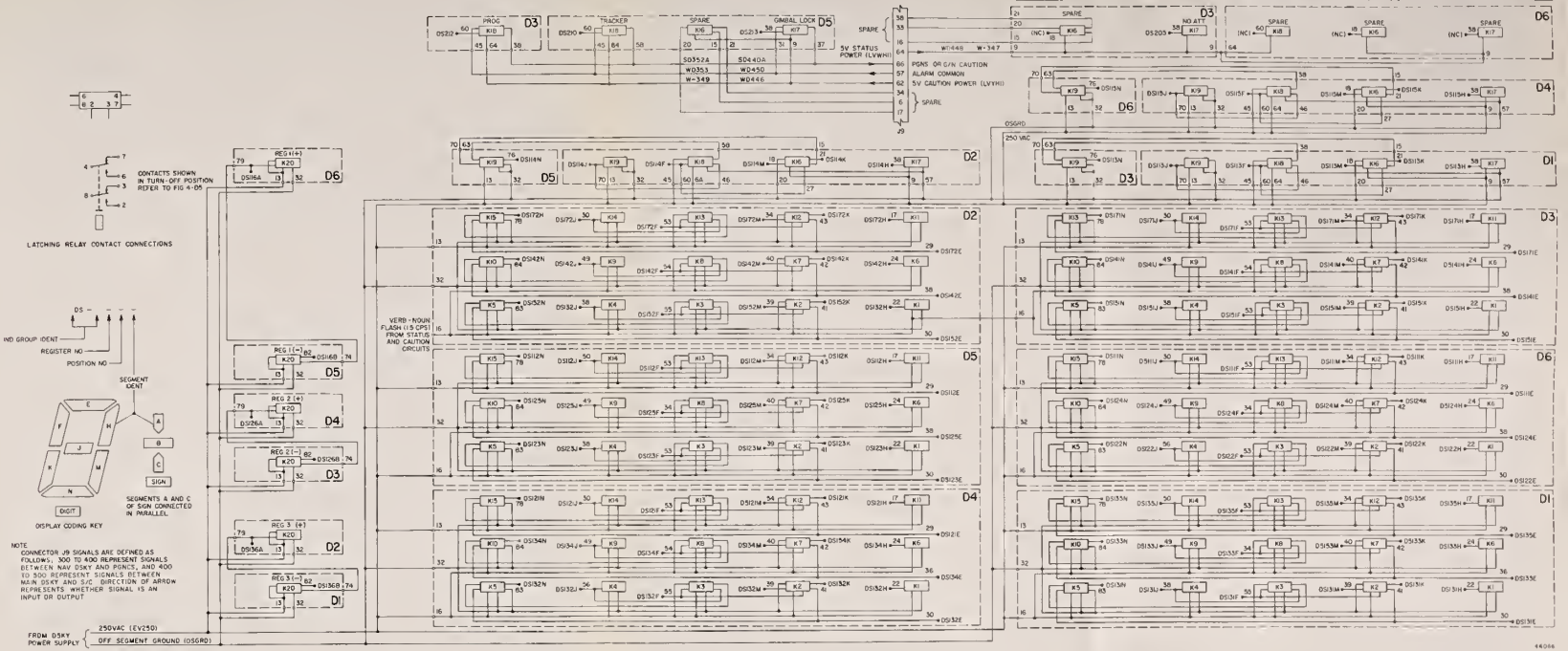


Figure 4-229. Relay Matrix Signal Flow Schematic Diagram

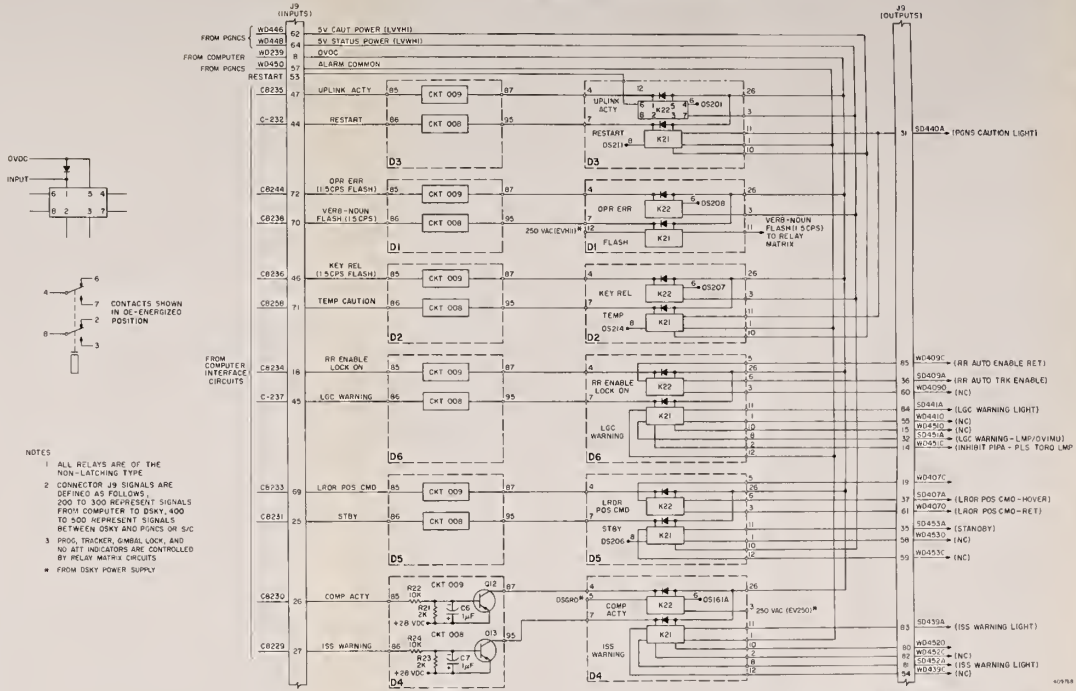


Figure 4-230. Status and Caution Circuit Schematic Diagram (LGC)

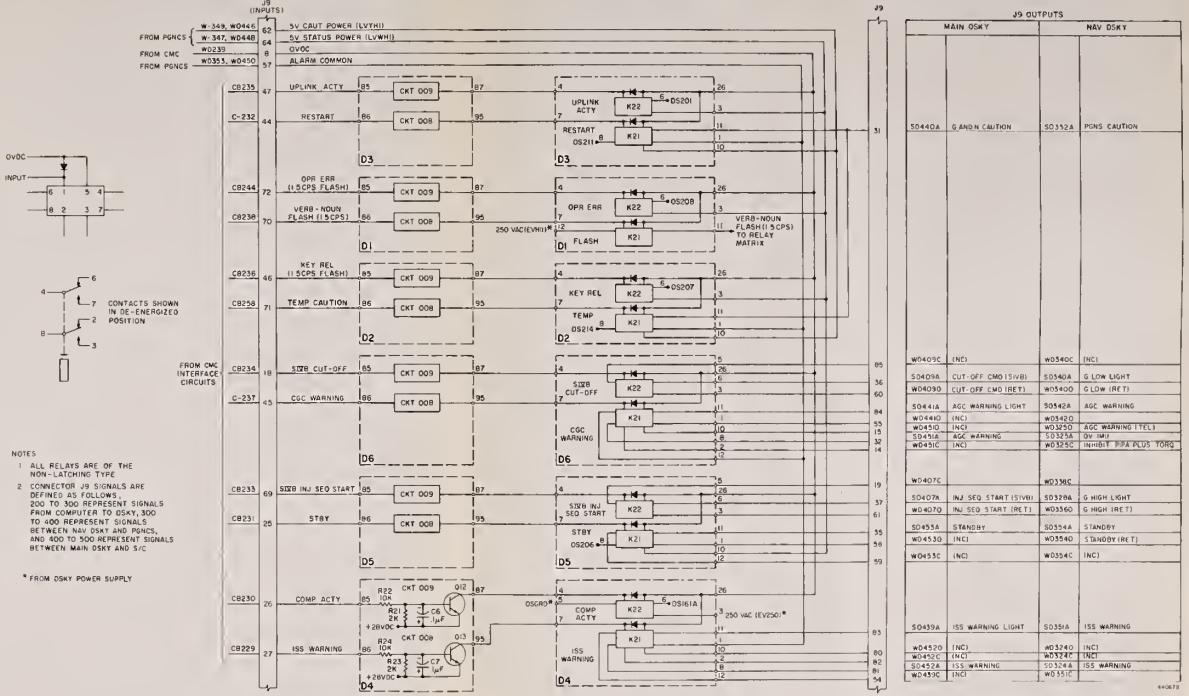


Figure 4-231. Status and Caution Circuit Schematic Diagram (CMC)